
AMBIENT
SYSTEMS
Reimagining the future of smart cities

40%

of global CO2 emissions is attributed to buildings

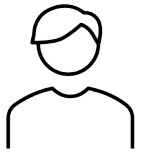
Climate Impact will largely be felt by building owners and their occupants

$260B+ of portfolio losses
(due to climate change)



Asset Manager





Occupant (Client)

Climate Risk can reduce asset value by 20% and annual returns by 40%

Ambitious Net Zero Emission Targets Set by COP UNFCCC (est. 1992) and enforced at national level

‘Conference of the Parties’

United Nations Framework Convention on Climate Change

May 1992-present

UN CLIMATE CHANGE CONFERENCE UK 2021

IN PARTNERSHIP WITH ITALY

31 Oct 2021 – 12 Nov 2021



13 Nov 2021



Growing list of Carbon Tracking Methodologies

PCAF Partnership for Carbon Accounting Financials

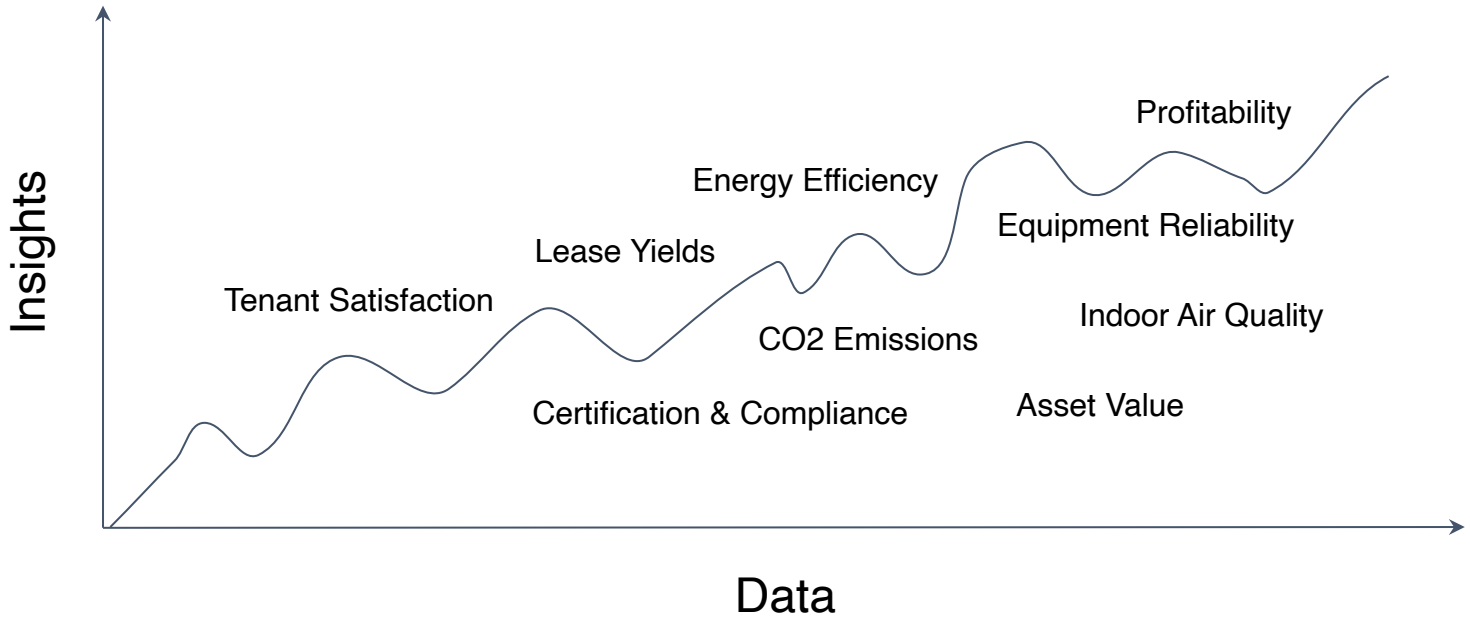
GREENHOUSE GAS PROTOCOL

SCIENCE BASED TARGETS
DRIVING AMBITIOUS CORPORATE CLIMATE ACTION

Global Reporting Initiative™

GRI GSSB Global Sustainability Standards Board

Data Science: empowering real estate asset managers to overcome climate risk



Identify Mispriced Assets
+
Operationalize Climate Resilience
+
Attract ‘Green’ Capital and Tenants

Status Quo

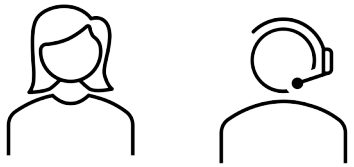
Current Challenges in CRE

Diverse



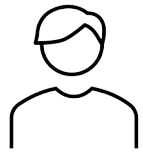
Asset Manager

Primary User



Facility or Property Manager

Key Stakeholder
Secondary User
Involved in Buying Process
Contributes Data



Occupant (Client)

End Beneficiary
Contributes Data

Gap Between Facility Managers and Property Managers

Facility Management

Driven by technical matters

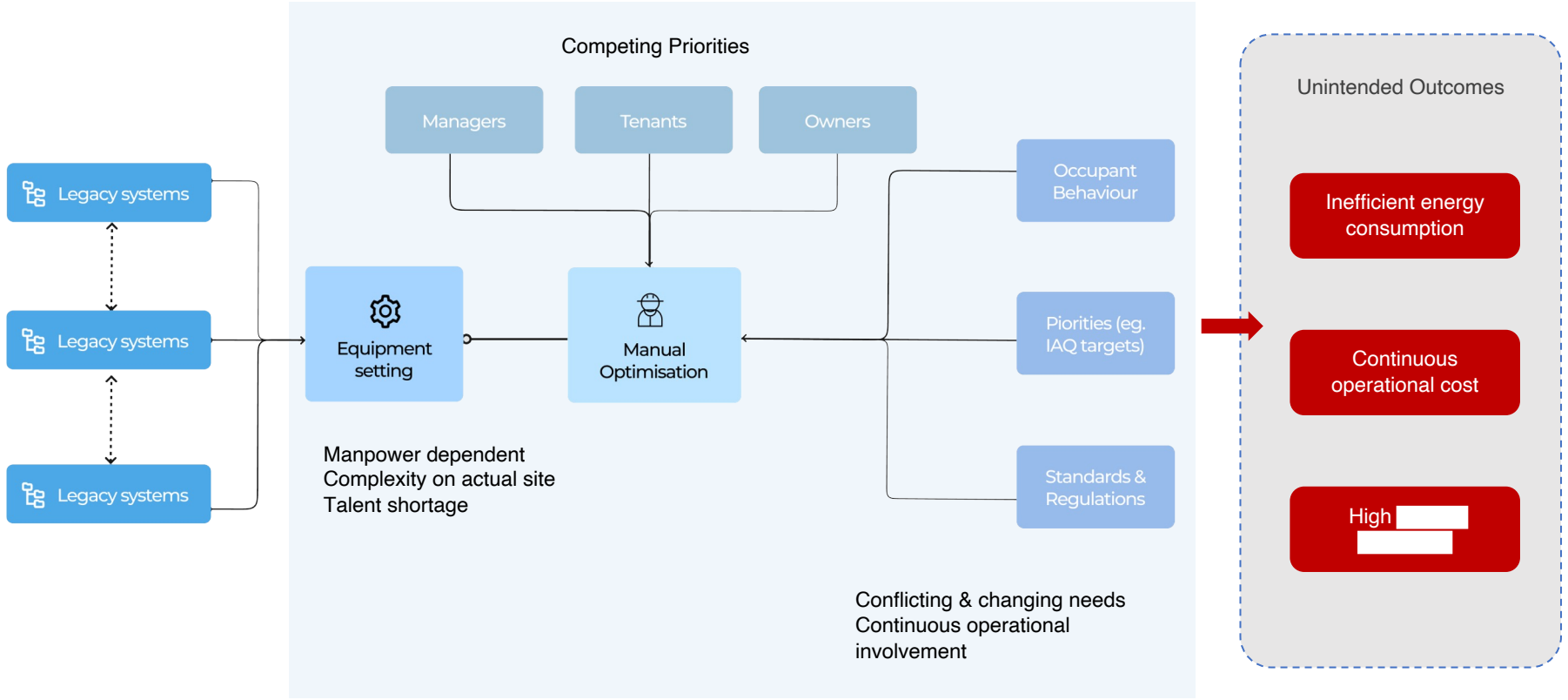
- Oversee and manage daily operations
  key facility management personnel
- Orchestrating maintenance & repair tasks
- Energy management
- Indoor Air Quality
- BMS / BAS operations

Property Management

Driven by financial matters

- Maxime property value and returns
  key personnel including property managers
- Work with leasing agents
- Communicating with investors and clients (occupiers)
- Limit risks & liabilities
- Ensure lease renewal

Decarbonization & Optimization in Buildings is Complex & Cumbersome



Hard to Find Expertise for Managing Net Zero Smart Buildings

Edge Devices

Automation

User Behavior

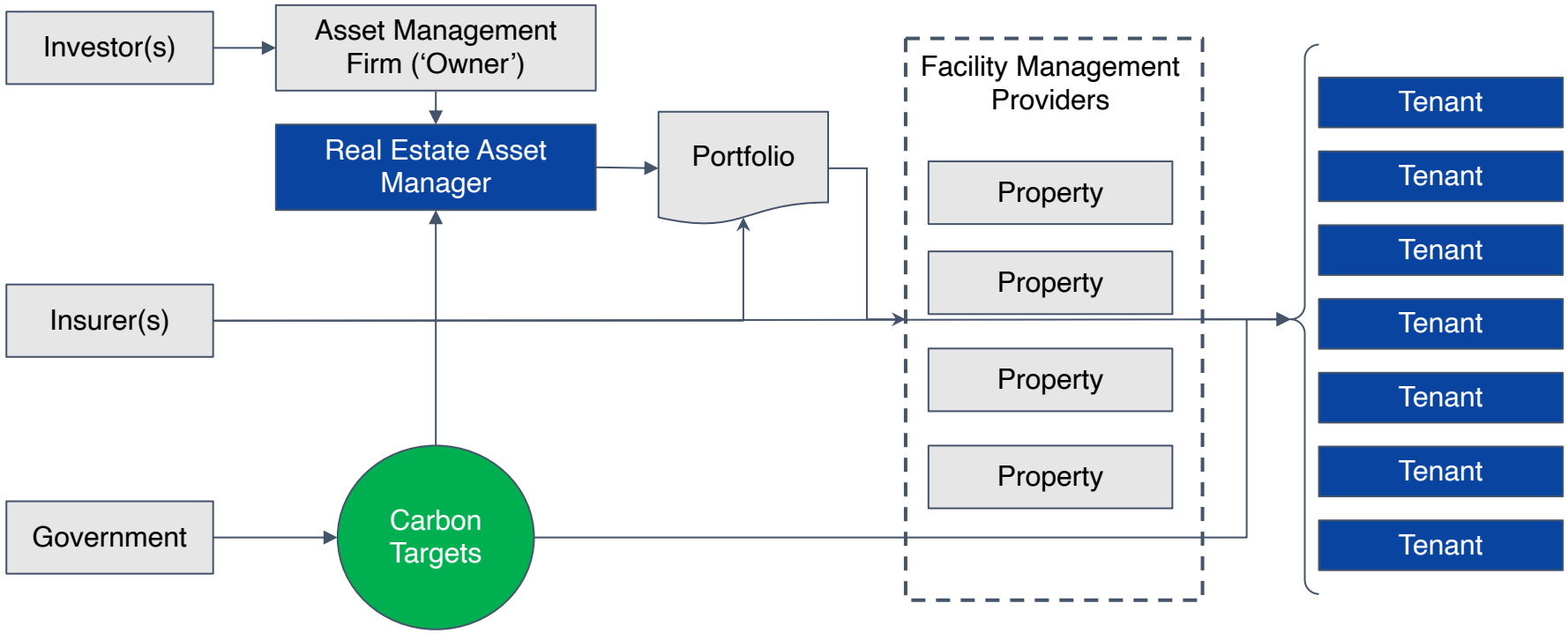


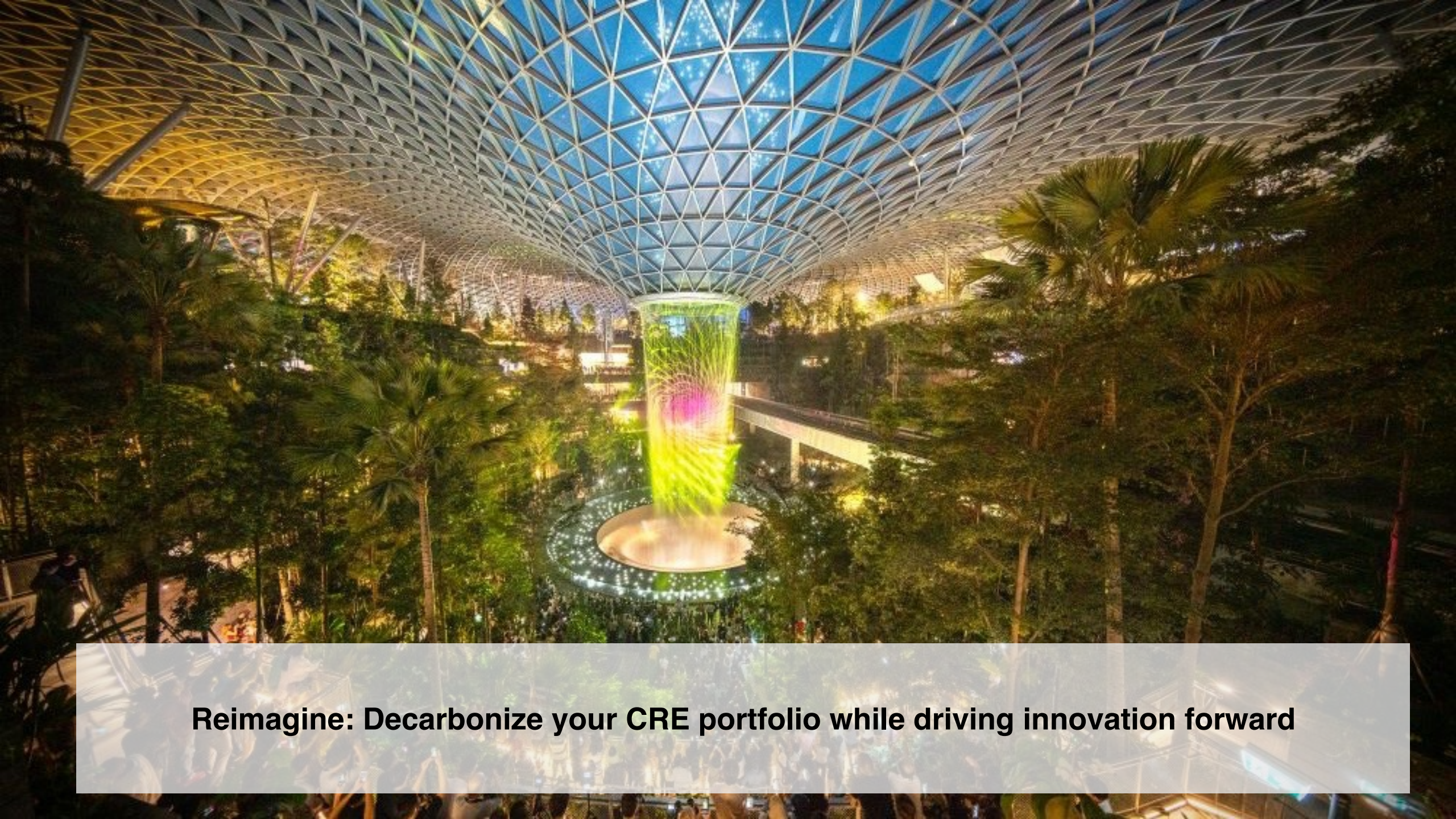
Inefficient energy consumption

Continuous operational cost

High

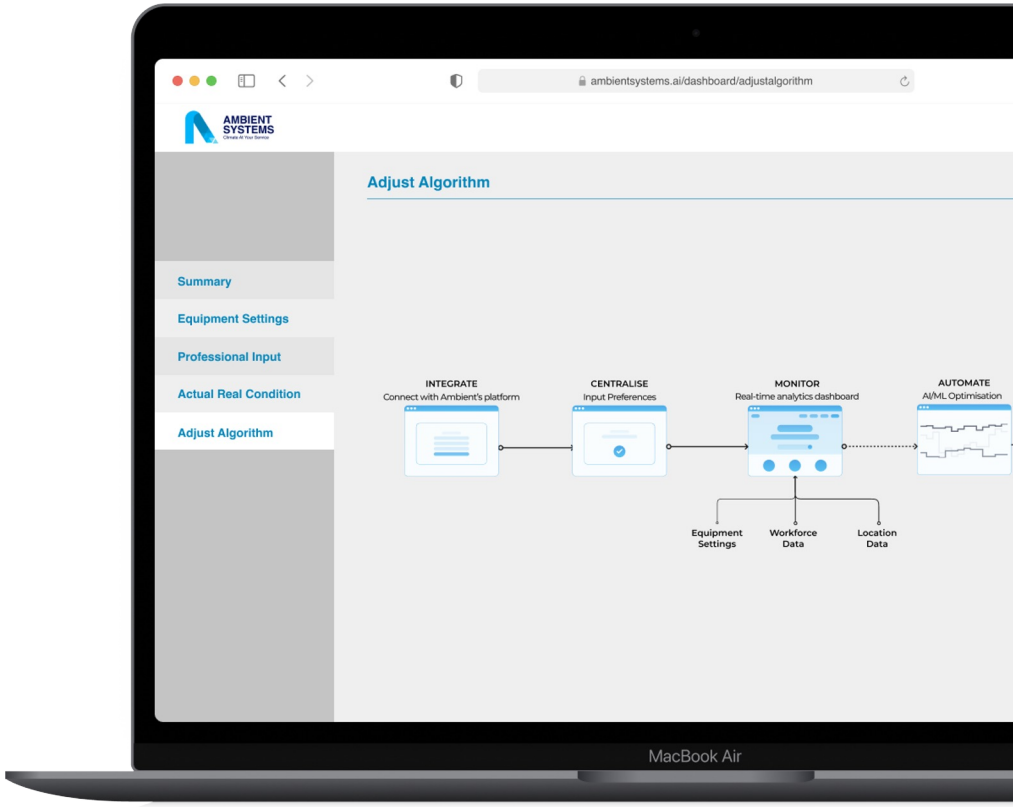
Complex Stakeholder Management Process





Reimagine: Decarbonize your CRE portfolio while driving innovation forward

Decarbonization Intelligence Platform for Commercial Real Estate



Supporting asset managers across the property lifecycle

15% to 25%

Enhanced Valuation Accuracy
(to reflect climate change risk)

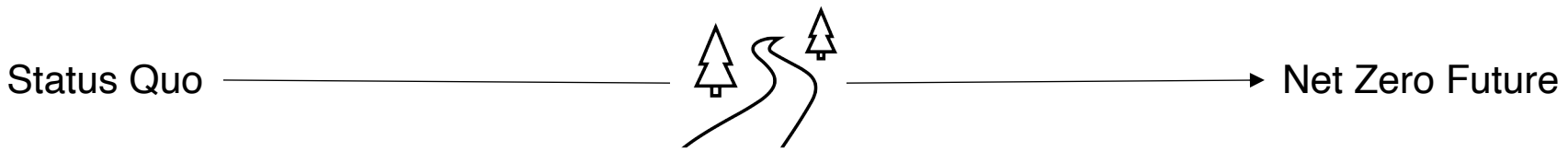
30%

Lower Operating Expense Ratio
(through operational efficiency insights)

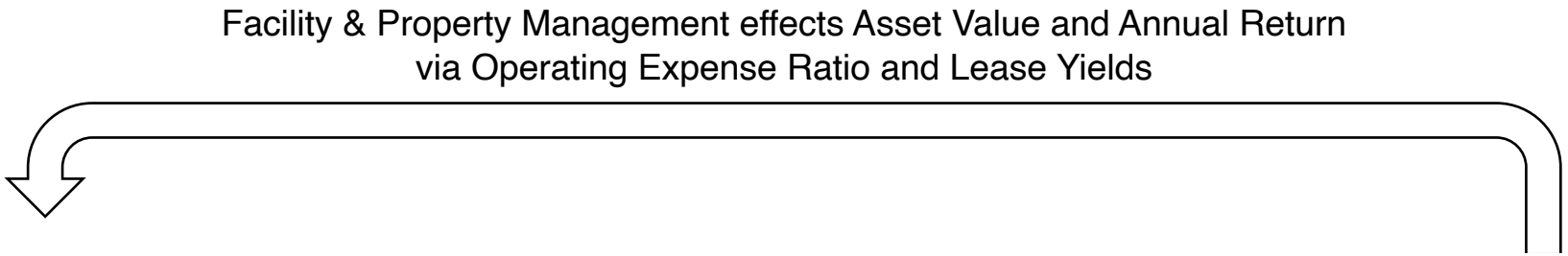
3X

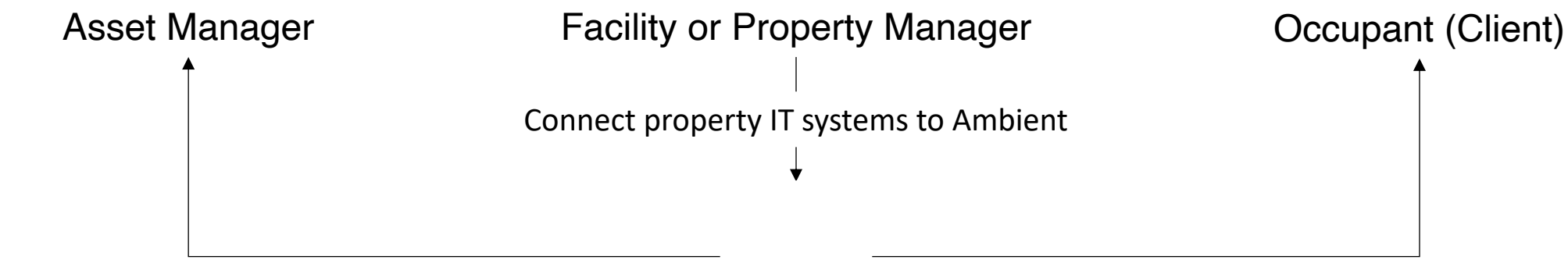
Modeling Time Savings
(compared to current process)

Moving Towards Decarbonization of Portfolio Assets & Occupant Operations



Ambient provides recommendations on how to improve property valuation & profitability





How exposed is this property to climate risk?
Where to invest to reduce emissions?
Which building systems need upgrades?
Which building technologies can help our tenants?

Which equipment is consuming most energy?
How can I use building technology to improve profitability?
Where is there room for upgrades?
How can I connect my production systems with building systems?

Balanced CRE optimization & decarbonization through three unique features



Flexible On-demand Subscription

No more onboarding, training and manpower reliance. Fast up-time and results through Ambient's subscription service



Simple and Affordable Pricing Plans

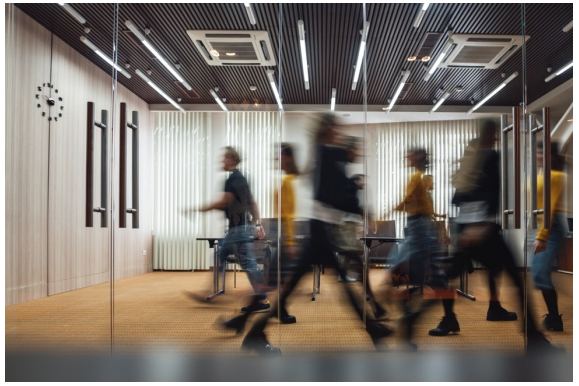
No more reliance on expensive consultants or in-house teams. One platform powered for all your data science use cases



Specialized AI Model Customization

Reach your business goals across energy efficiency, environment, and compliance with Ambient's state of the art AI models

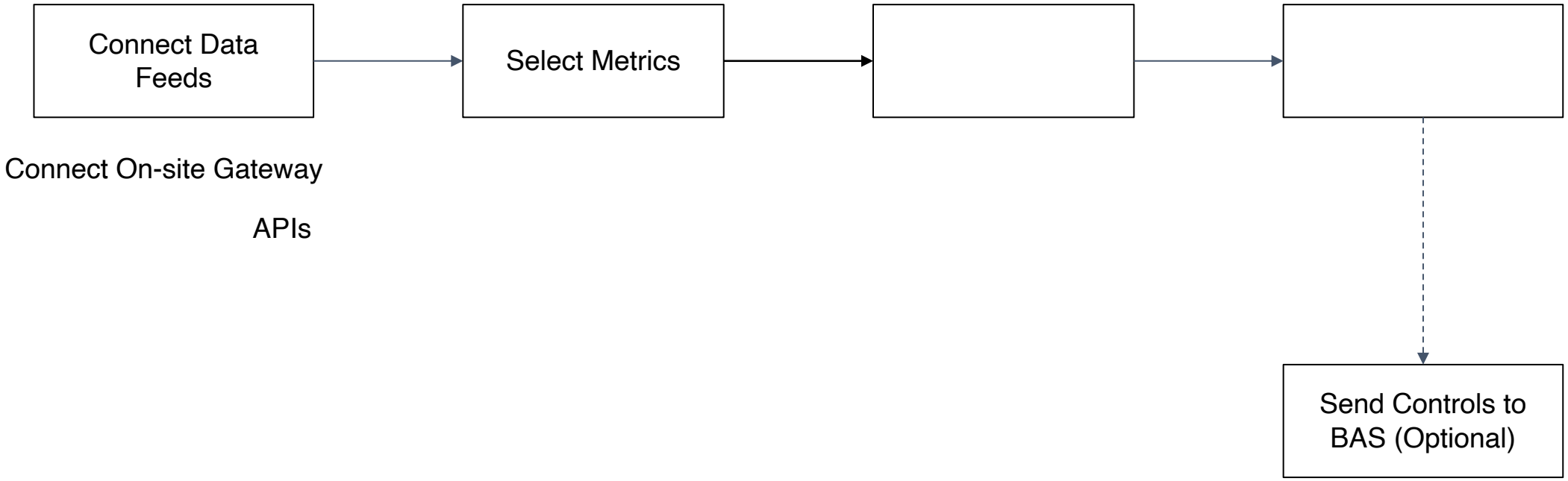
Ambient supports a broad a range of CRE assets

			
Industrial	**Retail**	**Offices**	**Transportation & Logistics**
Manufacturing R&D Life Sciences and BioTech	Malls Shopping Centers Grocery Stores	High Rise Low / Mid Rise Data Centers	Public Transit Airports Warehouses and DCs

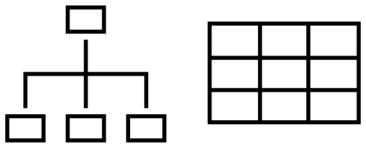
Ambient Systems

How it Works

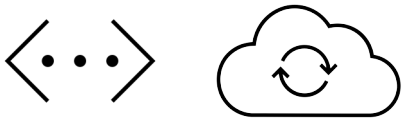
Platform: end-to-end CRE property optimization and decarbonization



Datasets: right datasets are key enablers of the machine learning models



Structured
Time Series Data
GPS Data
System Logs
CRM/ERP



Semi-Structured
Emails
HTML/XML files
JSON files
Network Traffic



Unstructured
Text
Audio
Video
Images

Variables: Ambient identifies mapping between key inputs and outputs

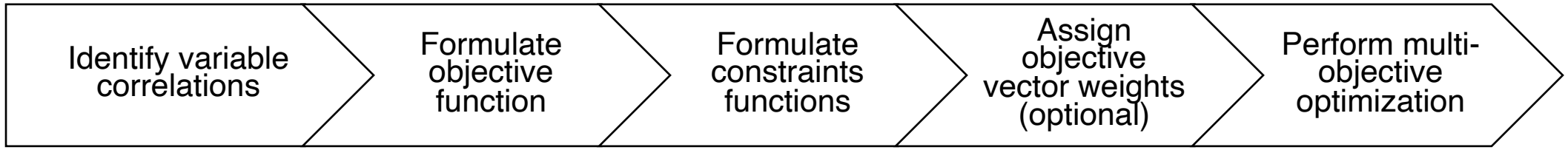
Independent Variables

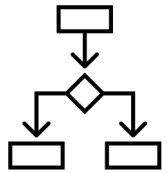
Local Supporting Infrastructure	**New**	**Ambient**
Building Technologies	**New**	
Climate Change Physical Risk	**Climate**	
Climate Change Transition Risk	**Climate**	
Employment Growth in Related Sectors	Conventional	
Employment Rate	Conventional	
Economic GDP Growth	Conventional	
New construction supply	Conventional	**Baseline**
Inflation	Conventional	
Facility Management Costs	Conventional	
Building Construction Type	Conventional	
Property Location	Conventional	

Dependent Variables

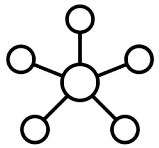
1. Asset Valuation & Appraisal Confidence
2. Occupancy / Vacancy
3. Operating Expense and Operating Expense Ratio

Models & Algorithms: latest machine learning advanced tailored for your portfolio





Expert Systems



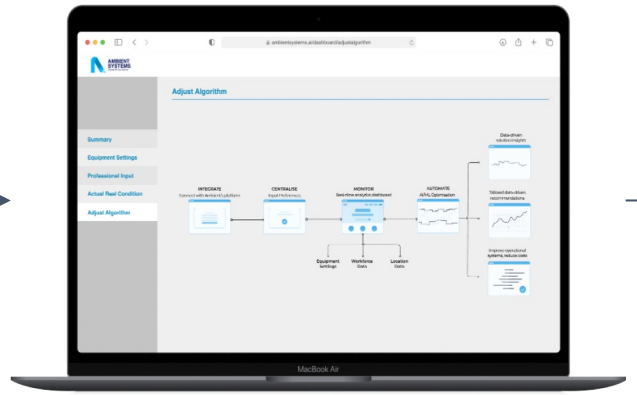
Knowledge Graphs



Stochastic Gradient Descent

Automation: Not a BMS or BAS but Compatible with your and Protocols

Ambient helps your facility managers make sense of BMS data and streamline



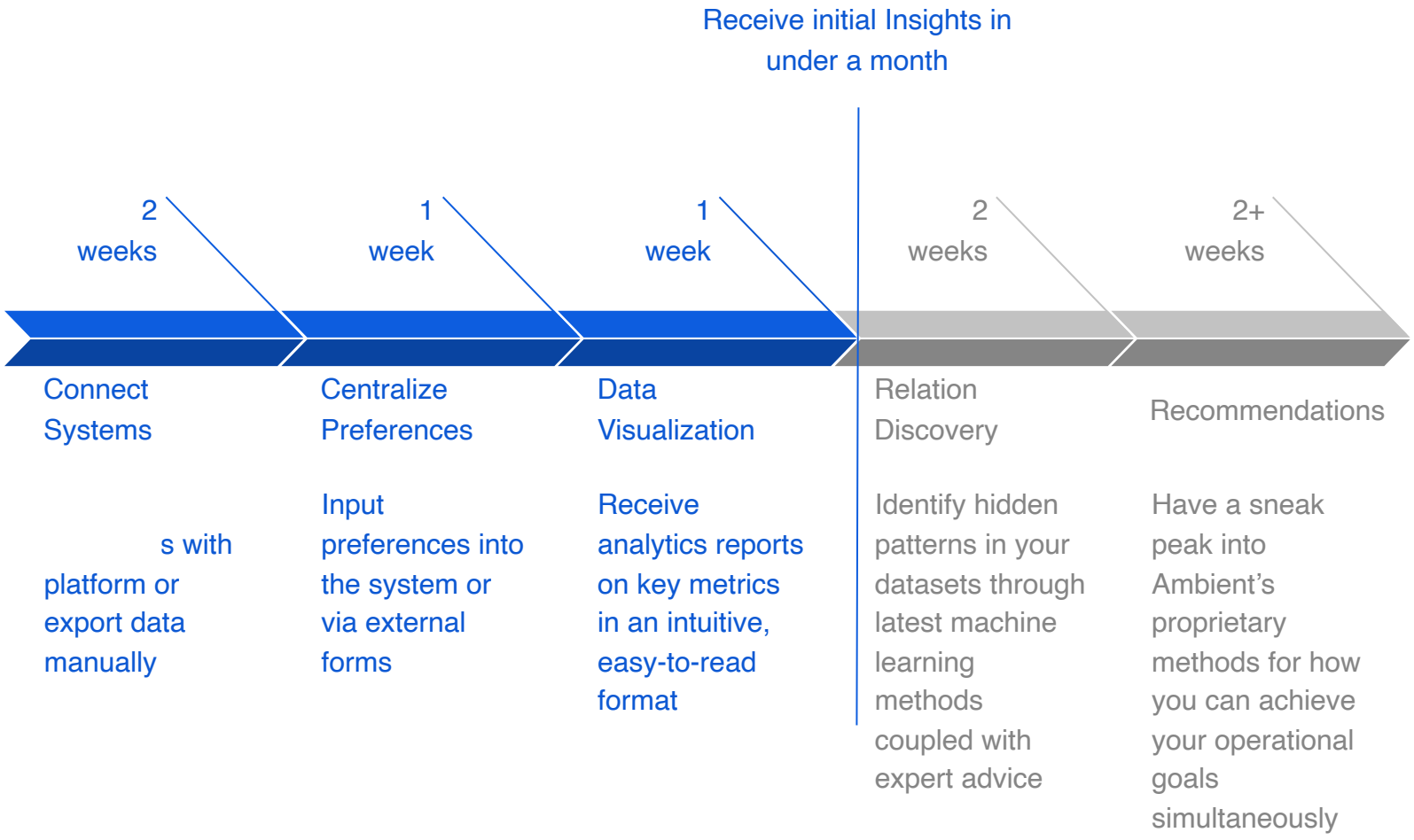
BMS Database

Siemens Building Technologies, , Honeywell, Bosch, ABB, Schneider Electric

Moving Forward

POC Project Proposal

POC duration can range 3 to 4 months with initial insights after 2 weeks from setting up the system



Simple steps towards effective POC execution and results

